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AIR MONITORING AND PURIFICATION

Solar Powered Algae-Based Air
Purifier for CO₂ Reduction

Name: Pari Chhabra

Registration No: 2427030599

Mentor: Dr. Sushama



INTRODUCTION

The Urgent Need for Sustainable Air Purification

Air pollution and rising CO₂ levels pose significant global challenges, driving climate change, global warming, and respiratory illnesses. Urban areas are particularly affected by high carbon emissions stemming from vehicles, industries, and fossil fuel consumption.

Limitations of Traditional Air Purifiers:

- High Electricity Consumption
- Only Filter Dust and Particles
- Ineffective at Reducing CO₂
- Not Environmentally Sustainable

Given these issues, there is a pressing demand for a sustainable, self-powered air purification system that effectively reduces carbon emissions.



A Streamlined Approach to Air Quality Monitoring and Improvement

Air pollution and escalating CO₂ levels pose significant environmental and health challenges. While Air Quality Index (AQI) values are tracked, most systems fail to pinpoint the specific causes of pollution in a given area.

Our strategy starts with the installation of air monitoring units in residential neighborhoods and schools. By employing a variety of sensor technologies—such as CO₂ sensors, gas sensors, temperature and humidity sensors, and particulate matter sensors—we will conduct a thorough analysis of the air quality in a targeted location. These sensors will help us identify the primary sources of pollution, whether they stem from vehicle emissions, industrial gases, elevated CO₂ levels, or particulate matter.

Once we have gathered and analyzed the data, we will introduce solar-powered algae air purifiers in the designated area. These purifiers will naturally absorb CO₂, enhancing air quality in an environmentally sustainable manner.



Literature Review

Existing air purification systems include:

- HEPA Filters: Effective for dust and particulate matter but unable to eliminate carbon dioxide (CO₂).
- Activated Carbon Filters: Absorb certain gases but have limited efficiency for all pollutants.
- Electrostatic Precipitators: Capture particles via electrostatic attraction but require high energy and costly maintenance.
- Limitations of Current Technologies:
 1. High energy consumption leading to inefficiency.
 2. Reliance on non-renewable energy sources, raising sustainability concerns.
 3. No solutions for biologically converting CO₂.

These limitations highlight the need for innovative air purification solutions that reduce environmental impact.



Algae-Based Carbon Capture Research

Research Insights:

- Microalgae can absorb CO₂ ****10 to 50 times faster**** than trees.
- Photosynthesis transforms CO₂ into ****oxygen and biomass****.
- Algae bioreactors are utilized in ****green buildings****.
- The integration of solar energy enhances ****sustainability****.

Challenges:

- Current systems are ****large-scale****.
- They tend to be ****expensive****.
- They are not designed for ****small public or household use****.

Identified Gap:

- 👉 There is a lack of a ****compact, affordable, solar-powered algae purifier**** suitable for urban settings.



Problem Statement

Addressing CO₂ Levels in Urban Environments

Urban areas are experiencing rising CO₂ levels as a result of swift industrialization and vehicle emissions. Current air purifiers primarily target particulate matter and do little to decrease carbon dioxide.

Furthermore:

- Traditional purifiers consume a large amount of electricity
- They indirectly contribute to an increased carbon footprint
- They are not sustainable over the long term

There is a pressing need for:

- A system powered by renewable energy
- A natural mechanism for absorbing CO₂

A compact and scalable solution



Proposed Solution

We present a Solar-Powered Algae Air Purifier that includes:

- A transparent algae chamber (bioreactor)
- A CO₂ intake pipe
- Solar panels
- An air circulation pump
- An oxygen outlet

A control system (with optional sensors)

##Before installation, a survey will be conducted to determine the sources of pollution in the area. This will be achieved using various sensors, such as the ESP32 gas sensor, humidity, and temperature sensors. This sensor technology survey is considered the most important part of the process.##



Working Principle

Step 1: Air with CO₂ enters through intake pipe

Step 2: Solar panels generate electricity

Step 3: Power runs air pump and circulation system

Step 4: CO₂ dissolves in water containing algae

Step 5: Algae perform photosynthesis

Reaction:

CO₂ + Sunlight → Oxygen + Biomass

Step 6: Oxygen-rich air is released



Sustainability Features

- ✓ Uses renewable solar energy
- ✓ Natural carbon capture
- ✓ Zero harmful emissions
- ✓ Produces oxygen
- ✓ Low operating cost
- ✓ Scalable design (home, office, public spaces)

Additional Benefits:

Biomass can be used for biofuel

Improves urban aesthetics

Educational environmental model



EXPECTED RESULTS

After implementation, the project aims for:

- Reduced localized CO₂ levels
- Continuous oxygen production
- Lower electricity consumption
- Sustainable air purification
- Increased awareness of biological carbon capture
- *Long-Term Impact:**
- Cleaner urban air
- Smaller carbon footprint
- Support for climate change mitigation
- Potential for smart city integration

The goal is to create a solar-powered, algae-based air purification system that targets CO₂ reduction and air quality improvement. It will use advanced sensors to monitor and address pollution in specific areas, promoting renewable energy and eco-friendly technology for urban air pollution solutions.



References

Journal & Research Based References

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*thank you
so much*

“Our solution does not just clean air
— it transforms pollution into life
using the power of nature and the
sun.”

